Letter Sequences & Alphabet Code (Set B)

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Spotting the Pattern

1. I — the gap between each letter is +2 (A, C, E, G, I).
2. V — the gap is -1 going backwards (Z, Y, X, W, V).
3. B (6) — F is the 6th letter of the alphabet.
4. A, B, C, D — C–E, G–I, M–O and P–R are all 2 apart; T and U are only 1 apart.
5. N — the gap is +3 throughout (B, E, H, K, N).
6. C (O) — O is the 15th letter of the alphabet.
7. Q — the gap is -2 throughout (Y, W, U, S, Q).
8. C — A, C, E, G, I (skipping one letter each time).
9. B (E) — shifting D forward by 1 gives E.
10. A, B, D, E — the gap is consistently 3, so the next letter after O is R, and the first letter given is C; the sequence moves forwards (not backwards), so C is false.
11. L — the gap is +2 throughout (D, F, H, J, L).
12. C (20) — T is the 20th letter of the alphabet.
13. I — the gap is -2 throughout (Q, O, M, K, I).
14. B (F) — shifting H back by 2 gives F.
15. A, C, E — P, R and W all come after M in the alphabet; J and L come before M.

Applying the Pattern

16. N — gaps alternate +2, +3, +2, +3: B→D(+2), D→G(+3), G→I(+2), I→L(+3), L→N(+2).
17. GEF — C+4=G, A+4=E, B+4=F.
18. S — gaps increase by 1 each time: A→D(+3), D→H(+4), H→M(+5), M→S(+6).
19. B (O) — 7+8=15, and the 15th letter is O.
20. A, B, D, E — all have a consistent gap of 4; C has a gap of 3.
21. N — the gap is consistently -3: Z, W, T, Q, N.
22. DFC — shifting back 3: G→D, I→F, F→C.
23. D — this interleaves A,B,C,D… (forwards) with Z,Y,X… (backwards): A, Z, B, Y, C, X, so the next term is D (continuing the forward sequence), with X being the paired backward term; accept D as the next single term shown.
24. B (T) — 26−6=20, and the 20th letter is T.
25. A, B, C, D, E — every pair is correctly shifted forward by 5, including Z→E which correctly wraps round past the end of the alphabet.
26. U — gaps increase by 1 each time: C→F(+3), F→J(+4), J→O(+5), O→U(+6).
27. Z — W is the 23rd letter; 23+3=26, which is Z.
28. Z — gap is -3 throughout: L, I, F, C, Z (wrapping: C is 3rd letter, 3−3=0, which wraps to 26=Z).
29. A, C, D — the gap is +3 throughout, E is the first letter, and the gap between K and N is 3, not 2, so E is false; the sequence is increasing, not decreasing, so B is false.
30. Z — S is the 19th letter; 19+7=26, which is Z.

Codes in Context

31. N — the gap is +3 throughout: B→E→H→K→N.
32. FMX — shifting BIT forward 4: B→F, I→M, T→X.

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33. B) N — the gap is −3 throughout: Z→W→T→Q, so Q−3=N.

34. 24 — C(3)+A(1)+T(20)=24.

35. The shift is +2 (B→D). Encoding PEN with +2: P→R, E→G, N→P, giving RGP.

36. B) P — the shift from E to H is +3, so M+3=P.

37. S — the gap is +4 throughout: C→G→K→O→S.

38. The shift is +3 (H→K). Decoding GRJ by −3: G→D, R→O, J→G, giving DOG.

39. 26 — D(4)+O(15)+G(7)=26.

40. A, B, D, E — the gap is 4 throughout (A), so the next letter is P+4=T (B); the gap between H and L is also 4 (D); D is the first letter (E). The sequence increases through the alphabet, so C is false.

41. M — the gap is −3 throughout: Y→V→S→P, so P−3=M.

42. LIT — decoding OLW by shifting back 3: O→L, L→I, W→T.

43. A) ECV — the shift from FOX to HQZ is +2, so CAT becomes ECV.

44. U — the gap is +4 throughout: E→I→M→Q→U.

45. D) 54 — S(19)+U(21)+N(14)=54.

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Homework Worksheet — Answers

Quick Practice

1. J — gap +2 throughout (B, D, F, H, J).

2. O — gap -2 throughout (W, U, S, Q, O).

3. C (11) — J is the 11th letter.

4. A, B, D — B–E, F–I and N–Q are each 3 apart; K–L and R–S are only 1 apart.

5. M — gap +3 throughout (A, D, G, J, M).

6. C (T) — T is the 20th letter.

7. M — gap -3 throughout (Y, V, S, P, M).

8. D — B, D, F, H, J (skipping one letter each time).

9. D (G) — shifting E forward by 2 gives G.

10. A, C, D — the gap is consistently 3 throughout (A); the next letter after P (16+3=19) is S (C); the first letter is D (D). The sequence increases, not decreases, so B is false; the gap between J and M is 3, not 4, so E is false.

11. M — gap +2 throughout (E, G, I, K, M).

12. A (13) — N is the 13th letter.

13. L — gap -2 throughout (T, R, P, N, L).

14. B (F) — shifting G back by 1 gives F.

15. A, C, E — G, C and F all come before J; M and L come after J.

Mixed Practice

16. O — gaps increase by 1: A→C(+2), C→F(+3), F→J(+4), J→O(+5).

17. GRJ — D+3=G, O+3=R, G+3=J.

18. T — gaps increase by 1: B→E(+3), E→I(+4), I→N(+5), N→T(+6).

19. C (T) — 9+10=19, and the 19th letter is T.

20. A, B, D — gap of 5 throughout; C has a gap of 5 then jumps to X (gap 6) — checking C,H,M,R,X: C(3)H(8)M(13)R(18)X(24), gaps are 5,5,5,6, so C is false; E has gap 4, so false. Correct: A, B, D.

21. M, J — gaps of -3: Y, V, S, P, M, J.

22. FMI — J−4=F, Q−4=M, M−4=I.

23. G — this interleaves A,C,E,G… (forward, skip one) with Y,W,U… (backward, skip one): A,Y,C,W,E,U,G — matching the pattern shown (A,Y,C,W,E,__), the next term in the forward sequence is G.

24. B (P) — 26−10=16, and the 16th letter is P.

25. A, B, C, D, E — all are correctly shifted forward by 6, including U→A and Z→F which both wrap correctly.

26. V — gaps increase by 1: D→G(+3), G→K(+4), K→P(+5), P→V(+6).

27. B — X is the 24th letter; 24+4=28, wraps to 28−26=2, which is B.

28. B — gap -3 throughout: N, K, H, E, B.

29. A, C, D — gap is +3 throughout and F is the first term; the next letter after R (+3) is U, so C is true; the sequence increases, not decreases, so B is false; gap between L and O is 3, not 2, so E is false. Correct: A, C, D.

30. T — N is the 14th letter; 14+6=20, which is T.

Standard / Advanced

31. S — the gap is +3 throughout; the next term after M is P, and the term after that is S.

32. E — the shift from FOX to HQZ is +2, so C becomes E.

33. I — the gap is −3 throughout; the next term after O is L, and the term after that is I.

34. C) E — the shift from HEN to KHQ is +3, so B becomes E.

35. G — the letter with position number 5 is E; 2 places after E is G.

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36. W — the gap is +4 throughout (C→G→K→O→S→W); two terms after O is W.

37. G — the gap is −3 throughout (V→S→P→M→J→G); two terms after M is G.

38. A) FDW — shifting CAT forward 3: C→F, A→D, T→W.

39. HKUJ — shifting FISH forward 2: F→H, I→K, S→U, H→J.

40. CAT — decoding ECV by shifting back 2: E→C, C→A, V→T.

41. A, B, E — the shift is +2 (D→F), so A is true; applying +2 to CAT gives E, C, V, i.e. ECV (B true, C false); SUN shifts to UWP (E true). D is false since the shift is +2, not −2.

42. O — the shift from C to F is +3, so L+3=O.

43. A) ECV — the shift from LEG to NGI is +2, so CAT becomes ECV.

44. A) LEST — the shift from BIRD to DKTF is +2, so decoding NGUV by −2 gives L, E, S, T.

45. FNC — shifting DOG back 1 gives CNF; swapping the first and last letters of CNF gives FNC.

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